Connecting to EC2

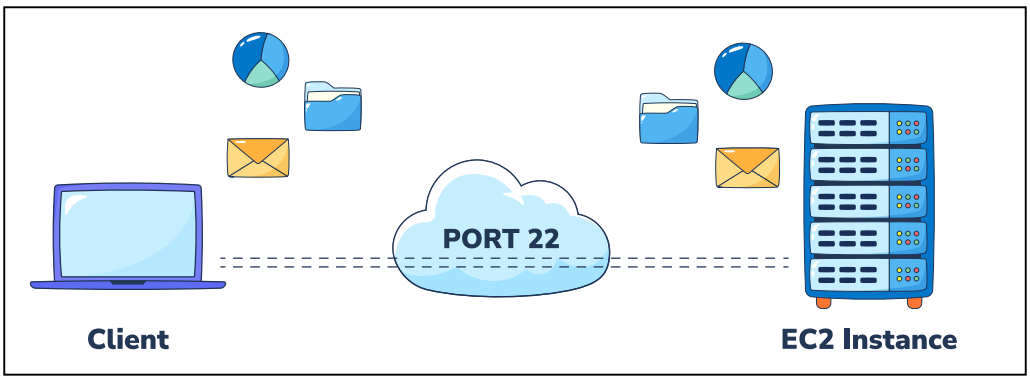
Explore different ways to connect to Amazon EC2 instances, including SSH for Linux and Remote Desktop for Windows. Learn how to use AWS Systems Manager Session Manager for secure, keyless access and best practices for maintaining connection security.

Connecting to Amazon EC2 instances is crucial to managing and working with virtual servers in the AWS cloud. Different methods for connecting to EC2 instances are available, including Secure Shell (SSH) for Linux instances, Remote Desktop for Windows instances, and AWS Systems Manager Session Manager's Instance Connect feature.

Let us take a deeper look at each of these methods:

Secure Shell for Linux instances

Secure Shell (SSH) is a cryptographic network protocol used for secure communication. It is commonly used to provide a secure and encrypted connection between a client and a server, allowing users to access and manage remote systems securely. SSH is widely used in various applications, including remote command-line access, file transfer, and tunneling.



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Secure Shell connection

Setup and connection steps for SSH

SSH requires a key pair to create a connection with the EC2 instance. A **key pair** consists of a public and a private key, a set of credentials to prove identity when connecting to an EC2 instance. We can divide the SSH connection into three major parts as follows.

- **Key pair creation:** When launching an EC2 instance, we can specify a key pair during the instance creation process. This key pair consists of a public key added to the EC2 instance and a private key securely downloaded to the local machine.
- **Permission setting for private key:** Ensure that the downloaded private key file has the correct permissions to be executed. For example, use `chmod 400` to restrict access to the owner only.
- **Connect using SSH:** We now open a terminal on your local machine and use the `ssh` command to connect to the EC2 instance using the private key:

```
ssh -i </path/to/private-key.pem> ec2-user@<instance-ip-address>
```

Here, `/path/to/private-key.pem` is the path to your private key file, `ec2-user` is the user for the Linux distribution, and `instance-ip-address` is the public IP or DNS of your EC2 instance.

Remote Desktop for Windows instances

Remote Desktop Protocol (RDP), a network protocol offered by Microsoft, provides a graphical interface to connect to a Windows-based computer o secure network connection. RDP allows users to interact with a remote desktop as if they were sitting in front of it, accessing its desktop environment,

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applications, and resources. It facilitates secure communication between remotely connected machines over an encrypted channel.

RDP is normally used in a client-server architecture; it operates on TCP and uses port 3389 by default. RDP offers secure remote access to Windows-based EC2 instances.

RDP connection

Setup and connection steps for RDP

We can divide the RDP connection into three major parts, as follows.

- **Download and install the remote desktop client:** Download and install a remote desktop client on your local machine, such as Microsoft Remote Desktop for Windows and macOS.
- **Administrator password retrieval:** When launching a Windows EC2 instance, retrieve the administrator password by decrypting the Windows administrator password from the EC2 console using the key pair specified during instance creation. Upload the private key file or paste the contents of the private key and decrypt the password.
- **Connect using a remote desktop:** Open the remote desktop client. Enter the public IP or DNS of your Windows EC2 instance and provide the administrator username and password retrieved earlier.

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AWS Systems Manager Session Manager (Instance Connect)

AWS Session Manager allows us to securely connect to the EC2 instances without managing SSH keys or opening inbound ports. We can create a temporary, secure session directly from the AWS Console or CLI. This simplifies administration and improves security by eliminating the need for complex bastion hosts.

Setup and connection steps for Instance Connect

Now, let's look at the connection steps for the Instance Connect:

- **IAM role configuration:** We first need to ensure that the EC2 instance has an AWS Identity and Access Management (IAM) role with the necessary permissions for Systems Manager.
- **Session Manager installation:** We then need the AWS Systems Manager Agent (SSM Agent) installed and running on the EC2 instance. Most Amazon Machine Images (AMIs) come with the SSM Agent preinstalled.
- **Systems Manager console connection:** On the AWS Management Console, navigate to Systems Manager, go to "Session Manager," and select "Start Session" for the desired EC2 instance.
- **Browser-based SSH or remote desktop access:** Finally, we can use the browser-based AWS Management Console to securely access the EC2 instance without external SSH keys or remote desktop client software.

Best practices for a secure connection

Establishing and maintaining secure connections to the EC2 instance is critical for protecting our data and infrastructure. Adhering to the following best practi help minimize unauthorized access and strengthen your overall security environment:

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- **Security groups:** Ensure that the security group associated with your EC2 instances only allows incoming traffic on the relevant ports, 22 for SSH and TCP 3389 for Remote Desktop.
- **Key pair management:** Keep your private key files secure and rotate them periodically for enhanced security.

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